

Volodya ALEKSANYAN

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EDUCATION AND DEGREES

Master's Degree in Mathematics and Artificial Intelligence Probability, optimization, advanced statistics, supervised and unsupervised learning, biostatistics (at MVA), deep learning, foundationals of ML, Time Series prediction MixtAI Scholarship (SaclAI-School): fellowship supporting diversity and academic excellence in AI	Université Paris-Saclay, Orsay 2025 – Present
Dual Bachelor's Degree in Computer Science and Mathematics <i>Mathematics:</i> probability, optimization, differential calculus, ordinary differential equations, linear algebra, statistics with R, Integration theory, numerical analysis <i>Computer Science:</i> fairness in AI, machine learning, distributed algorithms, theoretical computer science, logic, databases, functional programming, numerical optimization	Université Paris-Saclay, Orsay 2022 – 2025
Training on Gender-based and Sexual Violence	2023

RESEARCH PROJECTS & INTERNSHIPS

CVNN for Direct and Inverse Problems of EM Wave Diffraction <i>Python, Matlab - Supervisor Marc LAMBERT, laboratoire de GeePS, CentraleSupélec</i> Implementation of physics-informed complex-valued neural networks (CVNN) for improved resolution of direct and inverse electromagnetic wave diffraction problems	Mar 2026 – Present
Démonstration Interactive de l'Architecture Transformer <i>Python</i> Design of a web platform showcasing the attention mechanism in practice. Implementation of an emotion recognition module based on user input, with online training capability	Feb 2026 – Mar 2026
Prévision de la Demande Énergétique Durable <i>Python, R</i> Design of adaptive forecasting tools based on SARIMA, GAM, and XGBoost models to anticipate consumption variations driven by price volatility and renewable energy intermittency	Jan 2026 – Feb 2026
Bias Analysis in Vision-Language Models <i>Python - supervisor Alice HÉLIOU, Thales</i> Studied social biases in OpenAI's CLIP. Implemented and explored the impact of different fine-tuning, adversarial learning strategies on performance and fairness of CLIP applied to medical images	May 2025 – Jul 2025
Disease Prediction and Bias Mitigation on Chest X-rays <i>Python</i> Developed a fair classifier on a tabular data for predicting several pulmonary diseases. Increased the accuracy and minimized the biases of a deep learning model working with Chest X-Ray images	Feb 2025 – Apr 2025
Boolean Circuits <i>Python</i> Designed boolean circuits, automated circuit simplification and developed graph-based visualizations	Jan 2024 – May 2024

PROFESSIONAL EXPERIENCE AND ACTIVITIES

Member of Automatants CS IA - Artificial Intelligence Association of CentraleSupélec	Feb 2026 – Present
Ambassador Student - Help students in their integration, to inform them about the offers and services	Oct 2025 – Present
Student Monitor - Managed library resources, assisted students with research material	Sep 2023 – Jun 2025
Member of the Curriculum Advisory Board - Contributed to curriculum improvement and adaptation to professional requirements	Feb 2024 – Jun 2025
President, "LDD IM" Student Association - Strengthen student solidarity, facilitated networking with alumni and supported new student integration	Jun 2023 – Jun 2025
Co-founder and Vice-President of Algo Paris-Saclay Club - SWERC and Prologin preparation	Jan 2023 – Jun 2023

SKILLS & INTERESTS

Languages: French (native), Armenian (native), Russian (C2), English (C1)
Technical Skills: PyTorch, AIF360, Fairlearn, TensorFlow, Scikit-learn, Python, R, C, C++, OCaml, SQL
Interests: fairness in AI, algorithms, social engagement, cinemaphilia, long-distance running